SOC Phishing Email Detection

Project Overview

This project is a Security Operations Center (SOC) mini project that identifies phishing emails using Machine Learning. The system analyzes email text and predicts whether an email is phishing or safe.

Objective

To automate email classification and help security analysts detect phishing attempts quickly.

Features

- Email text preprocessing
- Feature extraction using CountVectorizer
- Email classification using Naive Bayes algorithm
- Accuracy evaluation
- Prediction of new emails

Technologies Used

- Python
- Pandas
- Scikit-learn
- CountVectorizer
- Multinomial Naive Bayes

Project Structure

```
soc-phishing-email-detection/
│
├── phishing_email_detector.py
├── README.md
├── requirements.txt
└── screenshots/
```

Installation

```
pip install pandas scikit-learn
```

Run the Project

```
python phishing_email_detector.py
```

Sample Output

```
Accuracy: 1.0

Confusion Matrix:
[[1 0]
 [0 1]]

Prediction: phishing
```

SOC Use Case

Phishing emails are one of the most common cyberattacks. This project demonstrates how machine learning can assist SOC analysts in detecting malicious emails and improving incident response.

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requirements.txt

```
pandas
scikit-learn
```